User-centered design process

Five phases, what each one produces, and where your project stands

- Neeta Khanuja

Reference project: a wayfinding tool for people with mild to moderate dementia

IMAMS Program: Bauhaus-Universität Weimar and University at Buffalo

USER-CENTRED DESIGN (UCD)

Design is a loop of five phases

1

Understand

Who are they, and what problems do they face?

2

Define

What exactly must our design do?

3

Ideate

What could we make?

4

Prototype

What will it look like, and how will it work?

5

Test

Does it work for the people we design for?



Iterate: what we learn in a test sends us back to an earlier phase.

Learn more: The 5 stages of design thinking and User-centered design (Interaction Design Foundation)

OUR REFERENCE PROJECT

Finding the way with dementia

Who: people with mild to moderate dementia, and their carers.

Problem: GPS maps can overwhelm them, and many stop walking alone.

Idea: directions built on landmarks the person already knows.



Master's thesis, Bauhaus-Universität Weimar, with the IDeA Center, University at Buffalo, 2019

PHASE 1 OF 5

Understand

Who are they, and what problems do they face?

What it means

We start with the people. We learn who they are, how they live, and what gets in their way. We also look at what already exists.

Why it matters

Designs built on guesses solve the wrong problem. A few conversations with real people prevent that.

OUTPUTS OF THIS PHASE

1. **Project brief**
   Who we design for, where, and their problem
2. **Existing solutions**
   What already exists, and what we can learn from it
3. **Interview and observation notes**
   What people say and what they do
4. **Insights**
   What we learned, each with its evidence

UNDERSTAND · 1 OF 4

Project brief

Who we design for, where, which problem, and what we leave out. It gives the team one shared target.

Project brief · Easy Walk

WHO

People with mild to moderate dementia who live at home, and their family carers

WHERE

Walking in their own neighbourhood

PROBLEM

Map apps overwhelm people with memory and orientation problems, so many stop walking alone

GOAL

Help them keep walking alone, guided by places they already know

OUT OF SCOPE

Driving, public transport, trips to unfamiliar places

LIMITS

Simple to learn; German rules on location sharing

Learn more: Frame your design challenge (IDEO Design Kit)

UNDERSTAND · 2 OF 4

Existing solutions

Products and services that already try to solve the problem. For each one, we note what it does and one lesson for our project.



NavMem

What it does: a phone app that gives walking directions step by step.

Lesson for us: tie each direction to a landmark.



Homing Compass

What it does: a small device that always points home.

Lesson for us: one simple function can be enough.



JAM Card

What it does: a card that asks staff for “just a minute” of patience.

Lesson for us: a discreet tool does not feel medical.



Walk Your City

What it does: street signs that show the walking time to nearby places.

Lesson for us: large text and one clear arrow.

How to review a solution: try it if you can · what does it do well? · what would not work for our users? · what will we borrow, and what will we avoid?

Learn more: Competitive usability evaluations (Nielsen Norman Group)

UNDERSTAND · 3 OF 4

Interview and observation notes

Raw notes from talking to people and watching them. They give real words and real behaviour.



OBSERVATION · WALK-ALONG

Who: Mrs. D., mild dementia, lives in a care home

Saw: plans her own afternoon walk to the flower shop; takes her walker and her watch; finds her way back

Said: she wears the watch so she is back before mealtime ends

INTERVIEW

Who: Avi, lives with a brain injury

“I never leave my apartment. I am too scared to leave my apartment.”

“When I get stressed out, I get confused.”

Note: he uses breathing exercises to calm himself and get back on track

What it contains: who · where · what they did · their exact words in quotes · a photo if they agree.

Learn more: User interviews 101 and Field studies (Nielsen Norman Group)

UNDERSTAND · 4 OF 4

Insights

Short statements of what we learned, each with its evidence. Truths the design must respect.

1 Many people with early dementia deny the condition.

Evidence: a common answer about the alarm armband was "I will use it when I need it."

2 Walks follow routines and time.

Evidence: Mrs. D. takes her watch so she is back before mealtime ends.

3 We remember places we have visited.

Evidence: "visited" was the most common reason people gave for remembering a landmark.

4 Fear can keep people indoors.

Evidence: Avi: "I never leave my apartment. I am too scared to leave my apartment."

How to get there: write each note on a sticky note, group similar notes, then write one sentence per group.

Learn more: Affinity diagramming (Nielsen Norman Group)

your project · 5 min

UNDERSTAND · YOUR PROJECT

Where does your project stand?

- [ ] **Project brief.** Can you say who, where and which problem in two sentences?
- [ ] **Existing solutions.** Which ones did you look at? What did each teach you?
- [ ] **Interview and observation notes.** Whom did you talk to or watch?
- [ ] **Insights.** Can you name one insight and its evidence?

Post in #in-class

1 Your project brief, in one or two sentences.

2 One insight, with its evidence.

3 One output you skipped. Why? Would it help now?

PHASE 2 OF 5

Define

What exactly must our design do?

What it means

We turn what we learned into a clear problem, then into the needs our design must meet.

Why it matters

Without clear needs, every idea looks equally good. The requirements at the end of this phase let us choose between ideas.

OUTPUTS OF THIS PHASE

1. **Problem statement and “How might we”**
   The problem in a few sentences, turned into questions
2. **Personas**
   Fictional people who stand for our users
3. **Journey map**
   The person’s experience today, step by step
4. **User stories**
   What each person needs, in their words
5. **Requirements**
   What the design must do. Everything above leads here.

DEFINE · 1 OF 5

Problem statement and “How might we”

A problem statement says who has the problem, what it is, and why it matters. “How might we” questions turn it into a starting point for ideas.

PROBLEM STATEMENT

People with mild to moderate dementia want to keep walking alone in their own neighborhood. Map apps give them too much information, and alarm devices feel medical, so many stop going out. Their family carers worry every time they leave the house.

How might we guide a walk using places the person already knows?

How might we let a carer know the person is safe, while the person keeps their freedom?

How might we make a tool that people want to carry, and that does not feel medical?

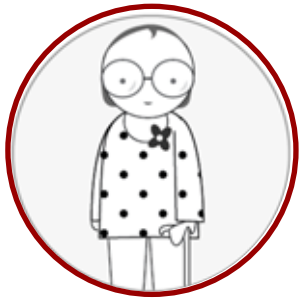
What it gives the team: one clear problem everyone agrees on, and questions that invite many answers.

Learn more: Problem statements (Nielsen Norman Group) and How might we (IDEO Design Kit)

DEFINE · 2 OF 5

Persona: Emma, 78

A persona is a fictional person who stands for one group of users. The **primary user** is the person who uses the design directly.



Emma, 78

Primary user

“I know my way. I just need a little reminder sometimes.”

ABOUT

- Has early dementia
- Lives alone in Arnsberg-Neheim, in the same neighbourhood for 30 years
- Her husband died 10 years ago
- Two children, three grandchildren
- Loves gardening, reading and music

GOALS

- Keep her daily walk
- Stay independent in her own home

FRUSTRATIONS

- New or changed routes confuse her
- Phone map apps show too much
- Refuses anything that looks like a medical aid

Learn more: Personas make users memorable (Nielsen Norman Group)

DEFINE · 2 OF 5

Persona: Ben, 45

A **secondary user** does not use the design every day, but supports the primary user or is affected by the design.



Ben, 45

Secondary user

“I want her to keep walking. I just want to know she got home.”

ABOUT

- Emma’s son, a doctor
- Married, with two children
- Lives in Old Arnsberg, 15 years
- He or his sister visits Emma every day

GOALS

- Know his mother is safe
- Respect her freedom to walk alone

FRUSTRATIONS

- Long work days; cannot check on her during a walk
- Emma refuses the assistive tools he suggests

Learn more: Personas make users memorable (Nielsen Norman Group)

DEFINE · 3 OF 5

Journey map: Emma’s walk today

A journey map follows one persona through one experience, step by step, as it happens today. It shows where things go wrong.

Persona: Emma · **Scenario:** an afternoon walk to the park · **Read it** left to right, stage by stage, and top to bottom.

STAGES	1. Decides to walk	2. Leaves home	3. Walks her usual route	4. Finds a street closed	5. Gets home late
ACTIONS	After lunch, plans a walk to the park	Takes her keys and coat; leaves her phone	Passes the church and the oak tree she likes	Takes another turn; nothing looks familiar	Finds her way back after an hour; Ben has called twice
THOUGHTS	“A walk will do me good.”	“I know this way.”	“There is my oak tree.”	“Where am I?”	“Maybe I should stay home next time.”
EMOTIONS	Happy	Confident	Calm	Anxious	Relieved, embarrassed
OPPORTUNITIES	Suggest the usual walk at the usual time	Make the tool easy to take along	Use the landmarks she already knows	Guide her back to a known landmark	Tell Ben she is home

What it gives the team: the moments that matter most. Stage 4 is the low point, so the design must help there.

Learn more: Journey mapping 101 (Nielsen Norman Group)

DEFINE · 4 OF 5

User stories

A user story is one need, written from the user's side: "As a [user], I want [something], so that [reason]."

S1

As **Emma**, I want **directions that name places I know**, so that I understand where to turn.

S2

As **Emma**, I want **one button that takes me home**, so that I can turn back whenever I feel unsure.

S3

As **Emma**, I want **a device that looks like an everyday object**, so that I feel comfortable carrying it.

S4

As **Ben**, I want **to set up Emma's usual places and routes**, so that the directions fit her life.

S5

As **Ben**, I want **a message when Emma asks for help**, so that I can call her right away.

S6

As **Ben**, I want **to know when Emma is back home**, so that I can stop worrying.

What it gives the team: small needs the team can discuss, rank and check off. The "so that" part keeps the reason in view.

Learn more: User stories with examples (Atlassian)

DEFINE · 5 OF 5

Requirements

Requirements say what the design must do. They are the goal of the Define phase. We sort them by priority.

Must have

Give one direction at a time, tied to a landmark she knows	S1
Have a physical Home button	S2
Have a Call button that alerts the carer	S5
Let the carer add places, routes and landmarks	S4

Should have

Look like an everyday object	S3
Tell the carer when the walk ends at home	S6
Ask the user before sharing location	HMW 2
Large text, high contrast	Insights

Could have

Suggest the usual walk at the usual time	Journey 1
Learn new landmarks from places she visits	Insight 3

What it gives the team: a checklist for judging ideas in the next phase, and for testing the prototype later.

Learn more: Product requirements (Atlassian)

your project · 5 min

DEFINE · YOUR PROJECT

Where does your project stand?

- [ ] **Problem statement.** Can you state the problem in two or three sentences?
- [ ] **Personas.** Who is your primary user? Is there a secondary user?
- [ ] **Journey map.** What does your user's experience look like today?
- [ ] **User stories.** Can you write three "As a … I want … so that …" stories?
- [ ] **Requirements.** Which three things must your design do?

Post in #in-class

1 Your primary user, in one sentence.

2 Your top "How might we" question.

3 One requirement your design must meet.

Skipping a step is fine. Knowing why you skipped it is the point.

PHASE 3 OF 5

Ideate

What could we make?

What it means

We create many ideas, explore them with the people we design for, and choose one concept.

Why it matters

The first idea is rarely the best one. Many ideas, shaped by real users, give us a better one to choose.

OUTPUTS OF THIS PHASE

1. **Brainstorm**
   Many ideas, quickly, with no judging
2. **Co-design session**
   Users and carers help shape the ideas
3. **Concept**
   The chosen idea, as a rough sketch and one page

Everything here stays rough: words, sketches, cards. Screens come in the next phase.

IDEATE · 1 OF 3

Brainstorm

A brainstorm produces many ideas quickly. We write every idea down and judge nothing yet.

How might we guide a walk using places the person already knows?

Talking directions	Photo of the next landmark ●●●	Watch buzzes at each turn ●	Arrows painted on the pavement
Ask-a-neighbour button	Route printed on a card ●	Smart walking stick	A map of only her places ●●
Big HOME button ●●●	Shop owners as helpers	Family records the directions ●	Call Ben with one press ●●●

GROUND RULES

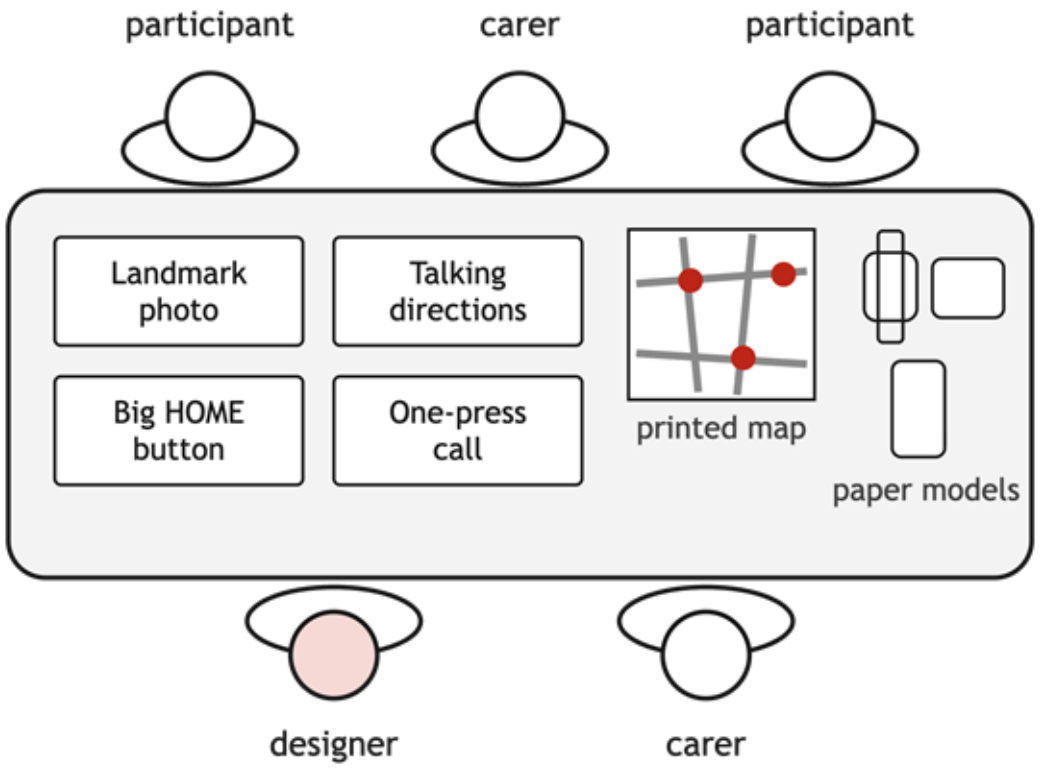
- Start from one “How might we” question
- One idea per sticky note
- Go for quantity; wild ideas are welcome
- Build on each other’s ideas
- At the end, vote with dots

Learn more: **Brainstorm** (IDEO Design Kit)

IDEATE · 2 OF 3

Co-design session

In a co-design session, the people we design for create with us. They sort ideas, add their own, and choose between options.



SESSION PLAN

Who: two people with mild dementia and their carers

1 Sort the idea cards: “I would use this” or “not for me”

2 Mark the map: the places you know and like on your walk

3 Pick a form: hold paper models of a watch, a handheld and a phone

WHAT CAME OUT

- The watch felt normal; a phone was “one more thing to forget”
- Favourite places made the best landmarks
- Carers wanted to know when a walk ends safely

Learn more: **Participatory design** (Interaction Design Foundation, video)

IDEATE · 3 OF 3

Concept

A concept describes the chosen idea on one page: what it is, who it is for, and how it works. It stays rough: words and sketches.

Easy Walk

In one line: walking directions built on landmarks Emma knows, on a simple wrist guide, set up by her family.

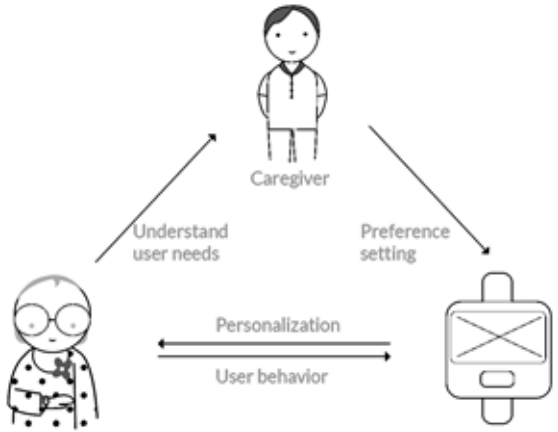
HOW IT WORKS

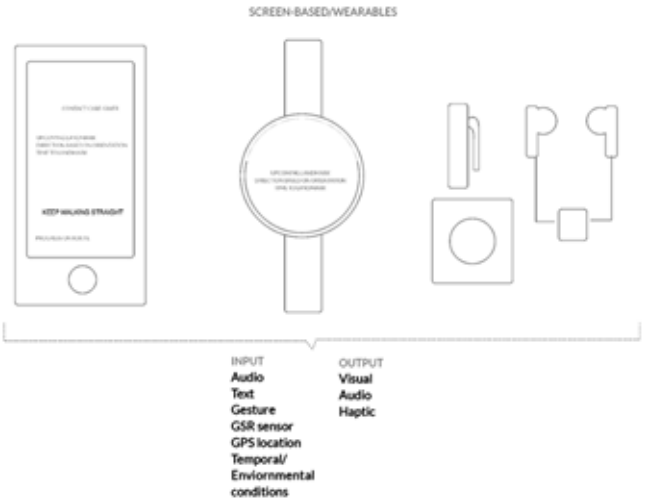
1 Ben adds Emma's places, routes and favourite landmarks in a phone app.

2 Emma wears a wrist guide. It shows one direction at a time, with the next landmark. Two buttons: Home and Call.

3 The tool learns new landmarks from the places Emma visits.

Meets: all four "must have" requirements.





Learn more: Create a concept (IDEO Design Kit)

your project · 5 min

IDEATE · YOUR PROJECT

Where does your project stand?

- [ ] **Brainstorm.** How many ideas did you have before choosing?
- [ ] **Co-design session.** Have users or carers shaped your ideas? Who could you invite?
- [ ] **Concept.** Can you describe your chosen idea on one page?

Post in #in-class

1 Your concept, in one sentence.

2 One idea you dropped, and why.

3 Who you could invite to a co-design session.

Skipping a step is fine. Knowing why you skipped it is the point.

PHASE 4 OF 5

Prototype

What will it look like, and how will it work?

What it means

We make the concept concrete, step by step, so people can see it and try it. We start rough and add detail.

Why it matters

People react to something they can see, hold or tap. Rough versions show problems early, when they are cheap to fix.

OUTPUTS OF THIS PHASE

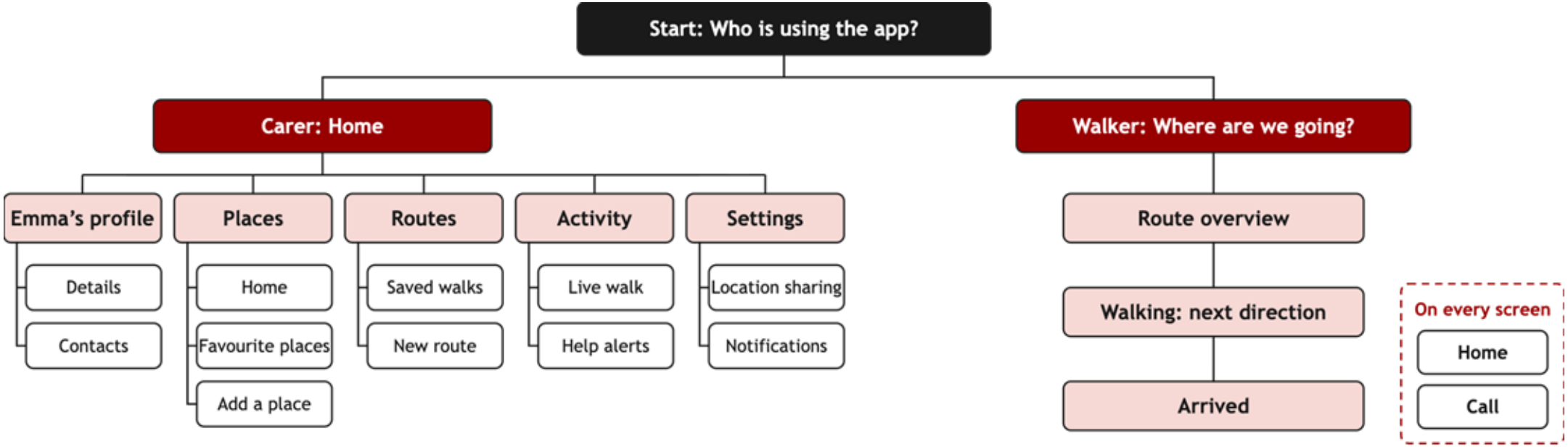
1. **Information architecture**
   Which pages exist, and how they connect
2. **Interaction flow**
   The steps a person takes to finish one task
3. **Wireframes**
   Simple sketches of each screen
4. **Storyboard**
   The design in use, drawn as a comic
5. **Low-fidelity prototype**
   A rough version people can try, often on paper
6. **High-fidelity prototype**
   Close to the finished product in look and feel

Fidelity means how close a prototype is to the finished product.

PROTOTYPE · 1 OF 6

Information architecture

Information architecture (IA) is the structure of a product: which pages exist, what they are called, and how they connect. It is often drawn as a sitemap.



What it gives the team: a map of every page, so the team knows what to design and how people move between pages.

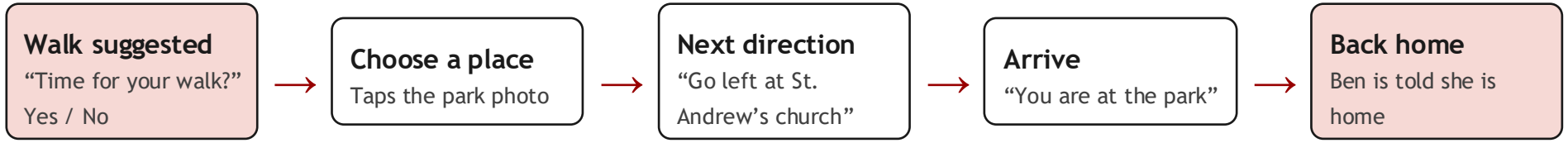
Learn more: Information architecture vs. navigation (Nielsen Norman Group)

PROTOTYPE · 2 OF 6

Interaction flow

An interaction flow shows the steps a person takes to finish one task, including choices and problems on the way.

Task: Emma walks to the park and back.



If she takes a wrong turn: “Turn around. St. Andrew’s church is behind you.” Then back to the next direction.

At any step: Home starts the route back. Call alerts Ben: “Call Emma. She is feeling lost.”

What it gives the team: every step and branch of a task, before anyone draws a screen. Each box later becomes a screen.

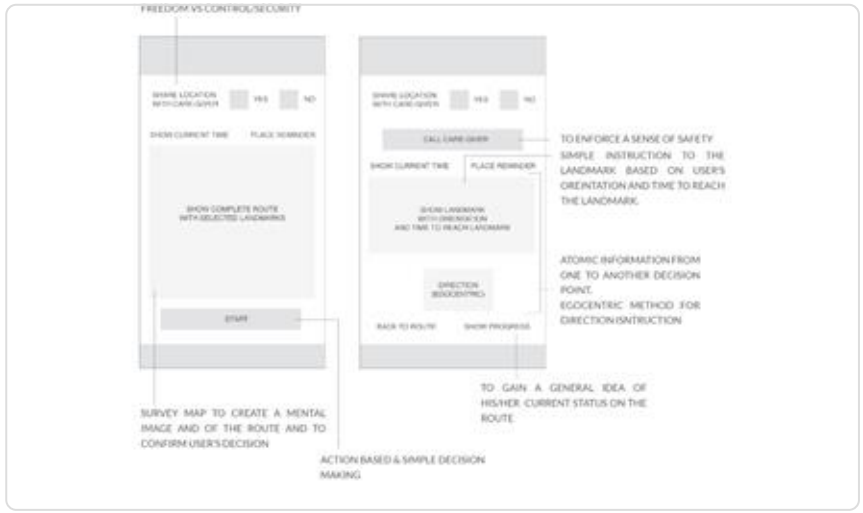
Learn more: User journeys vs. user flows (Nielsen Norman Group)

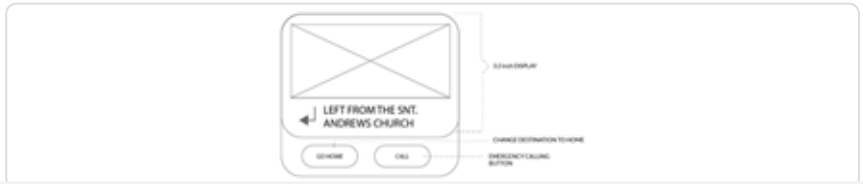
PROTOTYPE · 3 OF 6

Wireframes

A wireframe is a simple sketch of one screen. It shows what goes where, with grey boxes, real labels and no colour.







WHAT IT CONTAINS

Boxes for images and buttons

Real labels, such as “share location: yes / no”

Notes that say why: “To adapt to the user’s need, the tool has to know the user.”

No colour, so people judge the layout

What it gives the team: agreement on layout and content, quickly and cheaply, before any visual design.

Learn more: Wireframes (Interaction Design Foundation)

PROTOTYPE · 4 OF 6

Storyboard

A storyboard shows the design in use, as a short comic. Each panel is one moment, with the screen the person sees.



1 The wrist guide suggests her usual walk



2 With her permission, Ben is told she is out walking



3 One direction, one landmark



4 A wrong turn, gently turned back



5 She presses Call; Ben is alerted



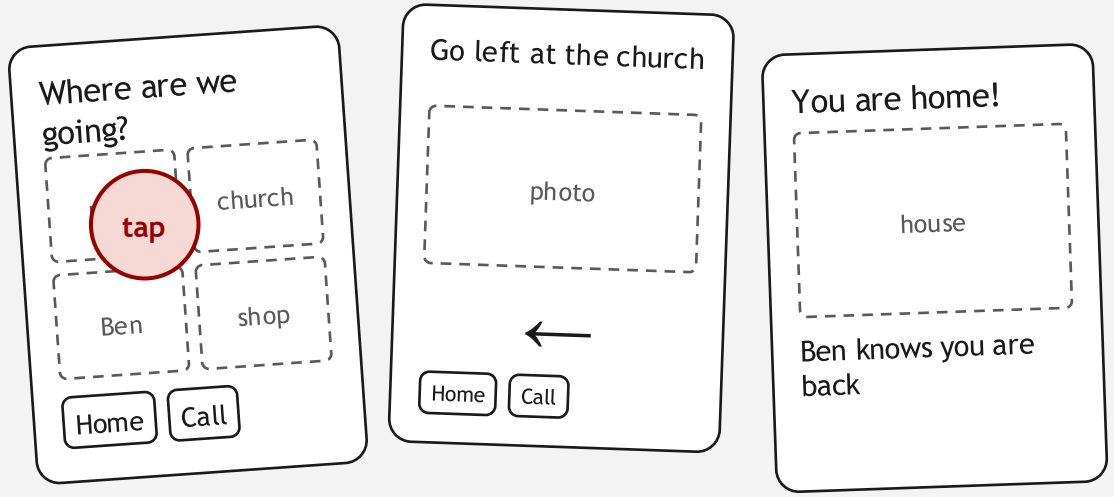
6 Home; Ben knows

Learn more: [Storyboards help visualize UX ideas](#) (Nielsen Norman Group)

PROTOTYPE · 5 OF 6

Low-fidelity prototype

A low-fidelity prototype is a rough version people can try. Paper is the most common material.



The wireframes, drawn on paper cards. A user taps a paper button; a team member swaps in the next card.

HOW TO MAKE ONE

1 One card per screen, from your wireframes

2 Give the user a task: “Walk to the park”

3 Swap cards as they tap; write down where they stop



For a physical product: a rough 3D print or cardboard model to hold, to judge size and buttons.

Learn more: Paper prototyping (Nielsen Norman Group)

PROTOTYPE · 6 OF 6

High-fidelity prototype

A high-fidelity prototype looks, and sometimes works, like the finished product. Each user gets their own interface.

Emma's interface

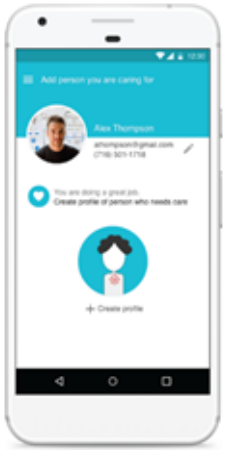


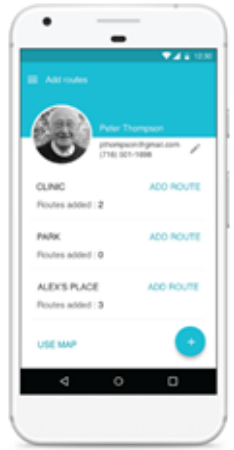


Wrist guide, and the walker side of the app: one direction at a time, Home and Call

Ben's interface







Carer side of the app: choose a role, add the person, add places and routes

Fidelity has two sides: how it looks and how it works. These look finished. Linked together in a tool like Figma, the screens become a clickable prototype to test.

Learn more: UX prototypes: low fidelity vs. high fidelity (Nielsen Norman Group)

your project · 5 min

PROTOTYPE · YOUR PROJECT

Where does your project stand?

- [ ] **Information architecture.** Can you list the main pages or parts?
- [ ] **Interaction flow.** Can you draw the steps for one key task?
- [ ] **Wireframes and storyboard.** Have you sketched the main screens, and the design in use?
- [ ] **Prototype.** What could people try this week? Paper is enough.

Post in #in-class

1 The one key task in your design.

2 The steps of that task, in a line.

3 What you will prototype first, and at what fidelity.

Skipping a step is fine. Knowing why you skipped it is the point.

PHASE 5 OF 5

Test

Does it work for the people we design for?

What it means

We watch real users try the prototype and write down what goes wrong.

Why it matters

We are not our users. Watching a few people try a prototype shows problems we cannot see ourselves.

OUTPUTS OF THIS PHASE

1. **Test plan**
   What we test, with whom, and how
2. **Findings**
   What went wrong, how serious it is, and what we change

Then we iterate. Each finding sends us back to an earlier phase. Design is a loop.

TEST · 1 OF 2

Test plan

A test plan answers six questions before we test. You can write it now, even if your test is weeks away.

1 What do we want to learn?	Can people with mild dementia follow Easy Walk on a walk they know?
2 Who will try it?	Five people with mild dementia, each with a family carer
3 What will they do?	Choose the park · follow the directions · press Home · press Call
4 What will we watch for?	Do they finish each task? Where do they stop or hesitate? What do they say?
5 Where?	In each person's own neighbourhood, on a route they know
6 How do we keep them safe?	Consent from both; the carer walks a few steps behind; stop at any time

What it gives the team: one agreed way to test, so every session runs the same and results can be compared.

Learn more: Checklist for planning usability studies and Usability testing 101 (Nielsen Norman Group)

TEST · 2 OF 2

Findings

Findings list what went wrong, the evidence, how serious it is, and what we change. These come from the project's early feedback.

Finding	Evidence	Severity	What we change
No help after a wrong turn	People want to know someone is watching	Switching between two roles in one app is complex	Non-working prototypes cannot be tested with people with dementia
"What happens if I take a wrong turn?"	*"Will someone know if I am off the route?"*	*Feedback on the first app*	*Expert review*
High	High	Medium	High
Add a "turn around" message with a landmark	Tell the carer when the person leaves the route	Split into a carer app and a simple device	Build a working prototype next
Useful to others who get lost			
"Something like this would be wonderful for my husband."			
Note			
Keep the design general enough			

What it gives the team: a ranked to-do list. Each finding sends us back to an earlier phase. This is iteration.

Learn more: Rating the severity of usability problems (Nielsen Norman Group)

your project · 5 min

TEST · YOUR PROJECT

Where does your project stand?

- [ ] **Test plan.** What will you test, and with whom?
- [ ] **Participants.** Can you reach people like your primary user?
- [ ] **Findings.** How will you record what goes wrong?

Post in #in-class

1 The one question your test must answer.

2 Who you will test with.

3 One task you will ask them to do.

Skipping a step is fine. Knowing why you skipped it is the point.

Learn more about each output

1 Understand

Frame your design challenge

Competitive usability evaluations

User interviews 101

Field studies

Affinity diagramming

2 Define

Problem statements

How might we

Personas

Journey mapping 101

User stories

Requirements

3 Ideate

Brainstorm

Participatory design (video)

Create a concept

4 Prototype

IA vs. navigation

User journeys vs. user flows

Wireframes

Storyboards

Paper prototyping

Low vs. high fidelity

5 Test

Usability test checklist

Usability testing 101

Rating severity

Iterative design

5 stages of design thinking

Sources: Nielsen Norman Group, IDEO Design Kit, Interaction Design Foundation, Atlassian. All free to read.

KEEP THIS

Your design checklist

1 Understand

- ☐ Project brief
- ☐ Existing solutions
- ☐ Interview and observation notes
- ☐ Insights

2 Define

- ☐ Problem statement and “How might we”
- ☐ Personas: primary and secondary user
- ☐ Journey map
- ☐ User stories
- ☐ Requirements

3 Ideate

- ☐ Brainstorm
- ☐ Co-design session
- ☐ Concept

4 Prototype

- ☐ Information architecture
- ☐ Interaction flow
- ☐ Wireframes
- ☐ Storyboard
- ☐ Low-fidelity prototype
- ☐ High-fidelity prototype

5 Test

- ☐ Test plan
- ☐ Findings

Download: Design checklist **(PDF, 6 pages)** · one page per phase, with space to write · **Thank you.**